Python Fundamentals (Assignment3)

Assignment Problems

Q1. Ask the user for a string and check whether it is a palindrome or not.

A **palindrome** is a string which is same when we read it forward & backward. Eg - "madam", "racecar" etc.

[**Hint** - A palindrome string is equal to the reversed version of the string. We can use a loop to reverse the string manually.]

Q2. Given a list of integers compute the average of all numbers in the list.

Q3. Input two lists of integers from the user. Merge them into one list and sort the result.

Eg - `list1 = [1, 2, 7]` , `list2 = [2, 4, 5]`

`result = [1, 2, 3, 54, 5, 7]`

Q4. Given a tuple of integers, create:

- A tuple of all even numbers
- A tuple of all odd numbers

manojghart691@gmail.com

Q5. Create a dictionary where:

- Keys = student names
- Values = marks (integer)

Write a menu-based program where user presses a key ('A', 'B', 'C', 'D') depending on the operation they want to perform on the dictionary:

1. `A` - Add a student
2. `B` - Update marks
3. `C` - Search for a student
4. `D` - Display all students and marks

Q6. Given a list of words:

```
words = ["apple", "banana", "kiwi", "cherry", "mango"]
```

Create a dictionary that maps each word to its length.

Example:

`{"apple": 5, "banana": 6, "kiwi": 4, ...}`

Q7. Write a program that takes a string from the user and prints the number of spaces in the string.

Q8. Write a program to check whether two lists share no common elements.

```
# share no common elements list1 = [1, 2, 3, 4] list2 = [5, 6, 7, 8]
# share common elements list1 = [1, 2, 3] list2 = [3, 4]
```

[**Hint** - use sets]

Q9. Given a list, print all elements that appear more than once in the list.

[**Hint** - use sets]

manojghart691@gmail.com

Q10. Ask the user for a string and print:

- All unique characters
- The count of unique characters

manojghart691@gmail.com

Assignment.py

```python
string = input("Enter any string: ")
n = len(string)-1
i = 0
value = True
while i <= len(string)//2:
    if string[i] != string[n-i]:
        value = False
    i += 1
if value == True:
    print("The string is palindrom.")
else:
    print("The string is not palindrom.")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter any string: manoj
The string is not palindrom.
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter any string: hrehoherh
The string is palindrom.
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

Assignment.py > ...

```python
from functools import reduce
listOfNumbers =[4,4,4,5,55,5]
avg = lambda x,y:(x+y)/2
avgOfnumbers = reduce(avg,listOfNumbers)
print(avgOfnumbers)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
17.375
PS C:\Users\USER\tryyyyyyyyyy>
```

C...

Code

Assignment.py

```python
list1 = [1,2,3,7]
list2 = [4,5,6]
list1.extend(list2)
mergelist = []
mergelist.extend(list1)
mergelist.sort()
print(mergelist)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
[1, 2, 3, 4, 5, 6, 7]
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

Assignment.py > ...

```python
number = (1,2,3,4,5,6,7,8,9,10)
oddtuple = tuple(i for i in number if i%2!=0)
eventuple = tuple(i for i in number if i%2==0)
print(number)
print(oddtuple)
print(eventuple)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
(1, 3, 5, 7, 9)
(2, 4, 6, 8, 10)
PS C:\Users\USER\tryyyyyyyyyy>
```

C...

Code

Assignment.py

```python
number = (1,2,3,4,5,6,7,8,9,10)
oddtuple = tuple(i for i in number if i%2!=0)
eventuple = tuple(i for i in number if i%2==0)
print(number)
print(oddtuple)
print(eventuple)

L = [1,2,3,4,5,6]
SL = tuple(i*i for i in L if i%2 == 0)
print(SL)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
(1, 3, 5, 7, 9)
(2, 4, 6, 8, 10)
(4, 16, 36)
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

Assignment.py > ...

```python
dic = {"Manoj":90,"Milan":85,"Krishna":88,"Hemraj":80,"Simanta":75,"Suresh":86}
while (1):
    option = input("Enter 'A' to Add new Student\nEnter 'B' to Update marks\nEnter 'C' to Search Student\nEnter 'D' to Display Students details\nEnter E to exits from the loop:\n").upper()
    match(option):
        case "A":
            name = input("Enter name of student: ").capitalize()
            while name in dic.keys():
                name = input("Enter another name of student: ").capitalize()
            mark = int(input("Enter mark of student: "))
            dic.update({name:mark})
            print("New data is updated.\n")

        case "B":
            name = input("Enter name of student whose mark do you want to update: ").capitalize()
            if name in dic.keys():
                new_mark = int(input("Enter mark of student: "))
                dic[name] = new_mark
                print("New mark is updated.\n")
            else:
                print("Not found!\n")

        case "C":
            S_name = input("Enter the name of student, do you want to search: ").capitalize()
            if S_name in dic.keys():
                print(f"{S_name}:{dic[S_name]}\n")
            else:
                print("Not found!\n")

        case "D":
            print("The name of students and thier marks")
            print(dic, "\n")
        case "E":
            print("Exiting from loop.")
            break
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter 'A' to Add new Student
Enter 'B' to Update marks
Enter 'C' to Search Student
Enter 'D' to Display Students details
Enter E to exits from the loop:
```

pow...

Code

Assignment.py

Assignment.py > ...

```python
words =["apple","banana","kiwi","cherry","mango"]
dic = {}
for i in range(len(words)):
    dic.update({words[i]:len(words[i])})
print(dic)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
{'apple': 5, 'banana': 6, 'kiwi': 4, 'cherry': 6, 'mango': 5}
PS C:\Users\USER\tryyyyyyyyyy>
```

pow...

Code

Assignment.py

Assignment.py > ...

```python
string = input("enter long string: ")
count = 0
for i in range(len(string)):
    if string[i] == " ":
        count+=1

print(f"The no of space present in string is {count}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
enter long string: My name is Manoj Gharti Magar.
The no of space present in string is 5
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

```python
list1 = list(input("Enter a list: "))
list2 = list(input("Enter another list: "))
print(list1)
print(list2)
value = True
for i in list1:
    if i in list2:
        value = True
    else:
        value = False

if value == True:
    print("Share common elements.")
else:
    print("Share no common elements.")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter a list: 1234
Enter another list: 45678
['1', '2', '3', '4']
['4', '5', '6', '7', '8']
Share common elements.
PS C:\Users\USER\tryyyyyyyyyy>
```

pow...
Code

Assignment.py

Assignment.py > ...

```python
given_list = [1,2,3,4,5,6,6,7,8,8,9,9,10,10,2]
new_list = list(set(given_list))
print(list)
print(new_list)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
<class 'list'>
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

Assignment.py > ...

```python
string = input("Enter the string : ")
unique_character = set(list(string))
print(unique_character,len(unique_character))
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter the string : manoj gharti magar
{'m', 'i', 'n', ' ', 'r', 't', 'o', 'a', 'h', 'g', 'j'} 11
PS C:\Users\USER\tryyyyyyyyyy>
```